作业纸

课程名称：大物II

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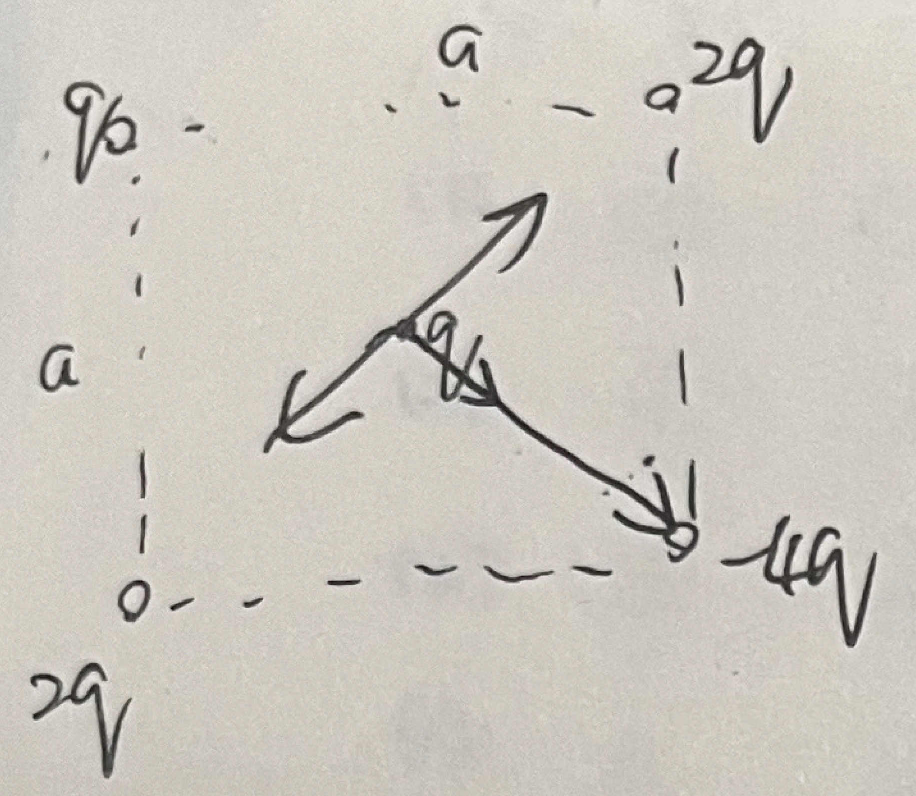
1-1.



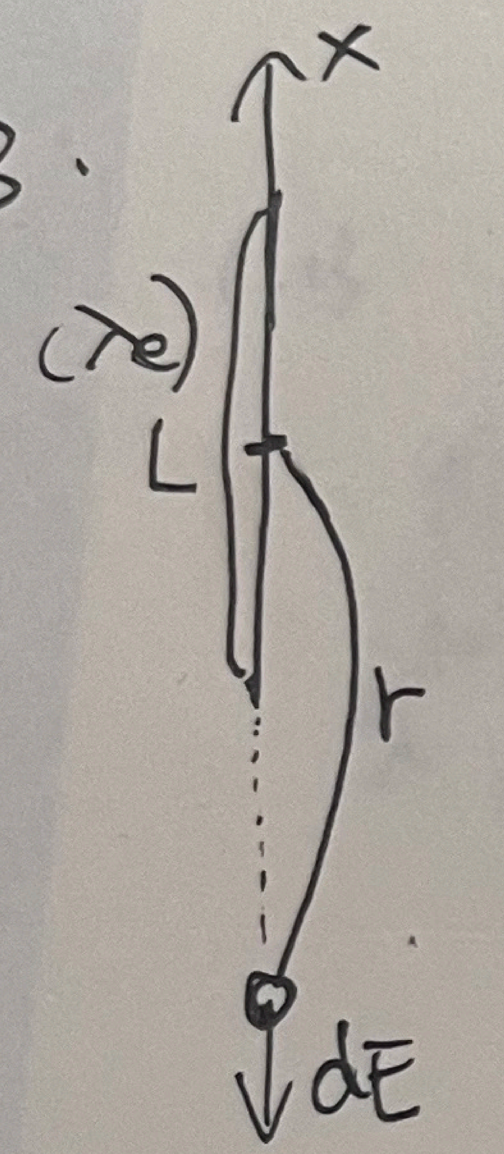
$r^2=\frac{1}{2}a^2$

$\vec{F}=\sum_{i=1}^{4}\frac{q_i q}{4\pi\varepsilon_0 r^2}\vec{e}_{r12}$

$=\frac{3q^2}{2\pi\varepsilon_0 a^2}\vec{e}$

$\vec{e}$ 方向指向-4q电荷方向.

1-3.



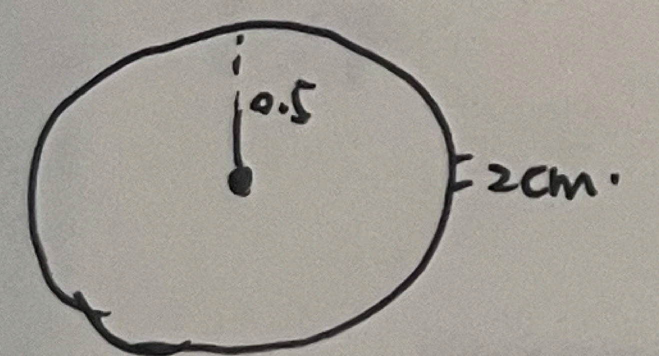
$dq=\lambda_e dx$.

$dE=\int_{r-\frac{L}{2}}^{r+\frac{L}{2}}\frac{dq}{4\pi\varepsilon_0 x^2}=\int_{r-\frac{L}{2}}^{r+\frac{L}{2}}\frac{\lambda_e dx}{4\pi\varepsilon_0 x^2}$

$=-\frac{\lambda_e}{4\pi\varepsilon_0 x}\Big|_{r-\frac{L}{2}}^{r+\frac{L}{2}}=\frac{\lambda_e}{4\pi\varepsilon_0(r-\frac{L}{2})}-\frac{\lambda_e}{4\pi\varepsilon_0(r+\frac{L}{2})}$

1-8.

相当于 2cm. 细棒的场强



$\lambda=\frac{q}{2\pi r-\cancel{\Delta x}}=\frac{3.12\times10^{-9}C}{3.12\ m}=10^{-9}\ C/m$

$q=\lambda\Delta x=10^{-9}\ C/m\times0.02m=2\times10^{-11}C$

$E=\frac{kq}{r^2}=\frac{9\times10^{9}N\cdot m^2/C^2\times2\times10^{-11}C}{\frac{1}{4}m^2}=0.72\ N/C$

联系方式 18835578460

1-11. $S = \pi r^2 = \frac{\pi}{400}\ m^2$

(1). $\Phi = ES\cos\theta = 300 \times \frac{\pi}{400} = \frac{3\pi}{4}\ N\cdot m^2/C$

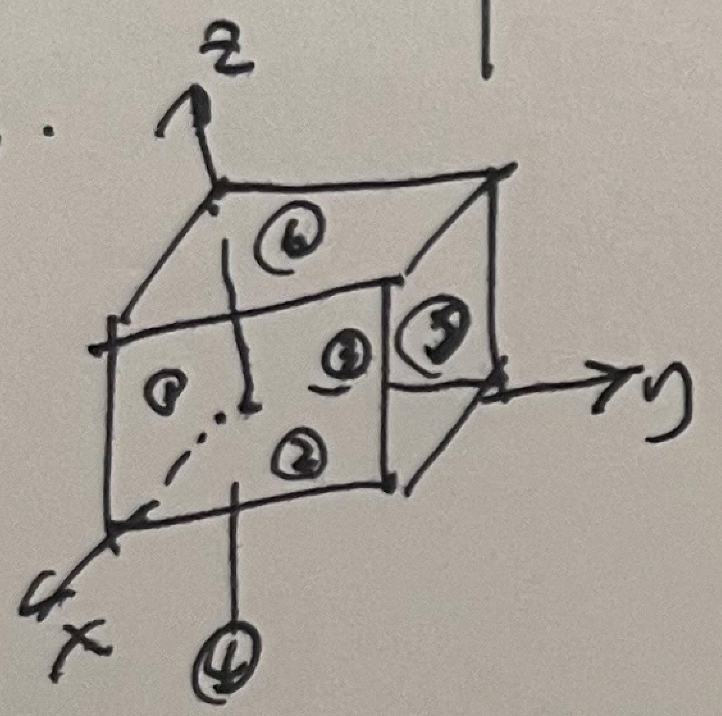
(2). $\Phi = \frac{3\pi}{4} \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}\pi}{8}\ N\cdot m^2/C$

(3). $\Phi = \frac{3\pi}{4} \times 0 = 0\ N\cdot m^2/C$

(4). $\Phi = \frac{3\pi}{4} \times -\frac{1}{2} = -\frac{3}{8}\pi\ N\cdot m^2/C$

(5). $\Phi = -\frac{3\pi}{4}\ N\cdot m^2/C$

1-13.



③ ④ $E = E_0(1+\frac{z}{a})\vec{i}$

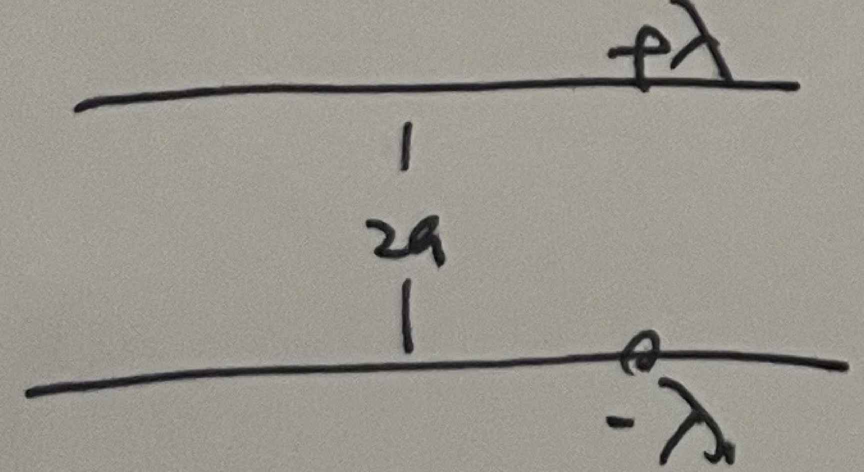
$\therefore \Phi_{③④} = 0$

② ⑥ $E_{②⑥} \neq 0$ $\Phi_{②⑥} = 0$

① ⑤ $E_{①⑤} = E_0(\frac{z}{a}) \neq 0$

$\Phi \neq 0$ ∴净电荷0

1-14



$\Phi = \frac{1}{\varepsilon_0}\sum q = \frac{\lambda L}{\varepsilon_0}$ $\Rightarrow \therefore E = \frac{\lambda}{4\pi\varepsilon_0 a^2}$

$\Phi = ES = EL\pi 4a^2$

$\frac{dF}{dL} = \frac{\lambda^2}{4\pi\varepsilon_0 a^2}$